

Precautionary Saving in Europe

Data, Pipeline, and Econometric Methodology

Project documentation

June 25, 2026

Abstract

This document explains, step by step, the empirical project behind the hypothesis that the euro area’s elevated household saving rate is *substantially precautionary* — a response to geopolitical and economic uncertainty rather than to income or wealth alone. It documents (i) the reproducible three-stage software pipeline, (ii) every data series used, with an explicit justification of its source, frequency, length, units, and treatment, and (iii) the econometric strategy in full detail: stationarity classification, the ARDL bounds test for a long-run relationship, Granger causality, and a structural VAR with orthogonalised impulse responses. A final section turns to the *composition* of saving — a “follow-the-money” test that, unlike the level analysis, can separate a yield-chasing reaction to ECB rates from a precautionary one, and finds the post-2022 reallocation is far more likely the former. Reported figures come from a clean end-to-end run; they are produced live and may be revised by the data providers, so they should be regenerated before citation.

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1 Overview and research question

Hypothesis. The euro-area household saving rate has stayed well above its pre-pandemic norm even after pandemic forced-saving and the income/wealth shocks unwound. We test whether this persistence is *precautionary*: driven by elevated uncertainty, it should

- (A) move *with* uncertainty over time (the time-series leg);
- (B) have risen *more* where the Russia/energy shock bit hardest (the cross-country leg);
- (C) be concentrated among households who can *afford* to save, while the fear itself weighs most on those who cannot (the distributional leg).

Each leg maps to a chart and, for leg (A), to a battery of time-series tests. The guiding methodological principle is that *co-movement is not causation*: uncertainty, policy rates, and inflation all rose together after 2022, so the descriptive charts can only show association. Section 5 is the attempt to isolate the precautionary channel formally.

What the documentation covers. Section 2 describes the reproducible pipeline. Section 3 justifies every data choice. Section 4 walks through the descriptive charts. Section 5 is the detailed econometric methodology and results. Sections 7–8 synthesise and qualify the findings. The appendix lists the file tree, exact commands, and a data dictionary.

2 The reproducible pipeline

The project is deliberately split into three scripts by *function*, so that each stage can be run, audited, and re-run independently. Data collection (slow, network-bound, non-deterministic across vintages) is separated from figure rendering (fast, deterministic, offline) and from econometric analysis (deterministic, offline).

Stage / script	Reads	Writes
01_collect_data.py	live APIs (Eurostat, FRED, GPR, OECD)	data/*.csv (tidy series)
02_make_figures.py	data/*.csv	figures/*.png (charts A–F)
03_econometrics.py	data/*.csv	data/econometrics_results.md, figures/A2*, A3*

One-command reproduction. The wrapper `run_all.sh` installs the pinned dependencies and runs the three stages in order:

```
bash run_all.sh      # pip install, then run 01 -> 02 -> 03 (classic Python)
```

Classic Python, no virtual environment. The project runs directly on a standard system Python (≥ 3.11); there is nothing to create or activate. `run_all.sh` installs the requirements into the chosen interpreter (python3 by default; override with the `PYTHON` variable) and then runs the three stages. Dependencies are pinned in `requirements.txt` so the numerical results are stable across machines. If the default python3 is a Homebrew/“externally-managed” build that refuses a global `pip install`, point `PYTHON` at a python.org interpreter or add `-user` to the pip line.

Network robustness. Downloads use the `requests` library, which ships its own certificate bundle (`certifi`); this avoids the `SSL: CERTIFICATE_VERIFY_FAILED` errors that the standard-library URL reader raises on a stock macOS Python. Each data pull is wrapped in `try/except`: a failed source prints a clear message and leaves any existing CSV from a previous run untouched, so a transient outage in one provider never corrupts the rest of the dataset. The Caldara–Iacoviello index is distributed as a legacy `.xls` file, which is why `xlrd` is a pinned dependency.

3 Data

This section is the heart of the “why this data” question. For each series we state the *source* and exact dataset code, the *frequency* and why it is appropriate, the *length/coverage*, the *units*, and any *transformation* and *caveat*. All series are free and require no API key. Spans below are from a representative run and will extend as providers release new vintages.

3.1 The target: euro-area household saving rate (quarterly)

Source / code

Eurostat sector accounts, NASQ_10_KI (key indicators), item SRG_S14_S15, sector S14_S15 (households and NPISH), seasonally and calendar adjusted (SCA), geography EA20.

Frequency

Quarterly. This is the natural and *binding* frequency: the saving rate is only published quarterly, and it is the dependent variable, so all higher-frequency regressors are aggregated down to it (Sec. 3.6) rather than the reverse.

Coverage

1999Q1–2025Q4 (≈ 108 quarters); median level $\approx 13.2\%$, latest $\approx 14.4\%$.

Units

Gross household saving as a percentage of gross disposable income (the gross saving rate). We use the *gross* rate because it is the headline Eurostat definition with the longest, most comparable series; this is noted whenever cross-source comparisons arise.

Why this exact series

The NASQ_10_KI table contains many ratios (saving rate, investment rate, profit share, ...) under different `na_item` codes. The collector *never averages across indicators*; it auto-selects the single dimension combination whose values look like a household saving rate—preferring a saving-named `na_item` and a level inside a plausible band of 5%–30%, closest to a typical ~13%—and refuses any candidate outside that band. This guards against a silent mis-selection that, in an earlier version, blended unrelated ratios into a nonsensical ~35% “saving rate”. The datasets NASA_10_KI and TEINA500 are tried as fallbacks.

3.2 Uncertainty proxies

Precautionary saving theory is about *perceived risk*, which is unobservable; we proxy it two ways and let the data adjudicate (Sec. 5.7).

Geopolitical Risk (GPR)

Caldara & Iacoviello, a newspaper-based index of geopolitical risk. Monthly, here from 1999-01 onward (the index itself extends back to 1900). It is the preferred proxy for this project because the framing is explicitly geopolitical (it spikes cleanly at the February 2022 invasion). Distributed as a keyless `.xls`.

EU Economic Policy Uncertainty (EPU)

FRED series EUEPUINDXM (Baker–Bloom–Davis methodology for Europe). Monthly, 1999-01 onward. A broader “economic-policy” notion of uncertainty, used as the robustness alternative to GPR.

Frequency rationale. Both are monthly; uncertainty can move sharply within a quarter, so a monthly source preserves the timing of spikes, which we then average to quarterly to match the target.

3.3 Confounders (controls)

Two variables co-move with uncertainty and independently affect saving; omitting them would bias any estimate of the precautionary channel.

ECB policy rate

FRED ECBDFR (deposit facility rate); fallbacks ECBMRRFR, then a 3-month interbank rate. Published *daily* (~10,000 observations from 1999); we take the quarterly mean. A higher policy rate raises the reward to saving, a competing explanation for the post-2022 rise.

HICP inflation

Eurostat PRC_HICP_MANR, all-items CP00, EA20, year-on-year %. Monthly, from 2000-12. High inflation both erodes real incomes and can itself raise precaution; it must be held constant.

3.4 Cross-country cross-section (leg B)

Saving by country

Eurostat TEC00131 (item SRG_S14_S15), *annual*, 21 EU countries, 2014–2025. Annual is the

only frequency at which a comparable cross-country household saving rate exists; for a cross-section that is sufficient.

Energy shock

Eurostat PRC_HICP_MANR restricted to energy (NRG), monthly YoY %. For each country we take the *peak* monthly energy inflation during calendar 2022, a single scalar measuring how hard the energy shock hit.

Derived join

`scatter_energy_vs_saving.csv` pairs each country's energy peak with its *change* in saving from 2019 to the 2023/24 average. Using the change (not the level) nets out structural North–South differences in saving habits, the more defensible test of the shock's effect.

3.5 Distribution (leg C)

There is no single official, annual, euro-area saving-rate-by-quintile series, so the project triangulates.

Eurostat ICW

ICW_SR_03 (experimental “Income, Consumption and Wealth” statistics), median saving rate by income quintile (QU1–QU5), unit PC_DI (% of disposable income), geography EA. Compiled only at multi-year reference points (~2010/2015/2020), so it is read as *structural* (the shape of the distribution), not as an annual path.

OECD EG DNA

Expert Group on Distributional National Accounts, attempted via the OECD SDMX API for an annual by-quintile panel. **Status:** the queried flow returns data without a usable quintile dimension, so chart E is not produced. The pull is retained, clearly flagged, for when the correct flow id is confirmed; it is a known data-availability limitation, not a code error.

ECB CES expectations

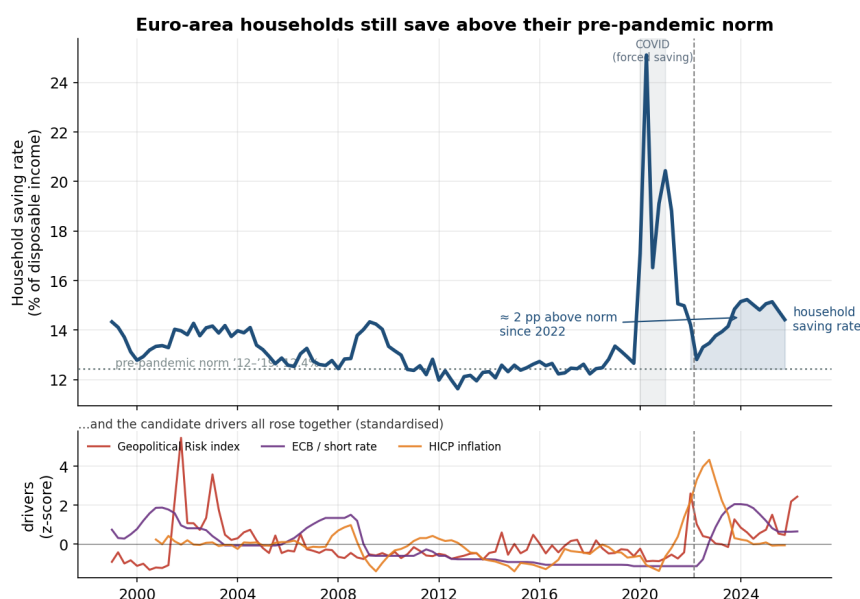
Expected unemployment 12 months ahead by income group, from published ECB Consumer Expectations Survey figures. These are not in any bulk API, so they are entered as documented constants in `01_collect_data.py` and written to `ces_unemp_expectations.csv`; refresh from the latest CES release before publishing.

3.6 Frequency alignment

The econometrics needs one common frequency. Because the dependent variable (the saving rate) is quarterly, *quarterly is the binding frequency*: every monthly or daily regressor is collapsed to a quarterly mean (`resample(QS).mean()`), and series are inner-joined on date. Aggregating down (averaging) rather than interpolating up avoids inventing within-quarter variation that the data do not contain. After alignment the estimation sample is **2000Q4–2025Q4 (101 quarters)**; the start is set by the shortest series (HICP inflation, from 2000-12) and the end by the target. The quarterly mean also lightly smooths the monthly uncertainty proxies, which is innocuous for the relationships we study.

4 Descriptive evidence

The charts are intentionally simple; they motivate the formal tests but do not substitute for them.



Top: saving level vs its 2012–19 norm. Bottom: drivers standardised to z-scores. They co-move, so co-movement alone can't isolate the precautionary channel — see 03_econometrics.py for the VAR / bounds test.

Figure 1: Leg A. Top: the euro-area household saving rate against its 2012–19 norm, with the post-2022 excess shaded and the February 2022 invasion marked. Bottom: the candidate drivers—uncertainty, the ECB rate, and HICP inflation—standardised to z-scores. They visibly rise together, which is precisely why eyeballing cannot isolate the precautionary channel.

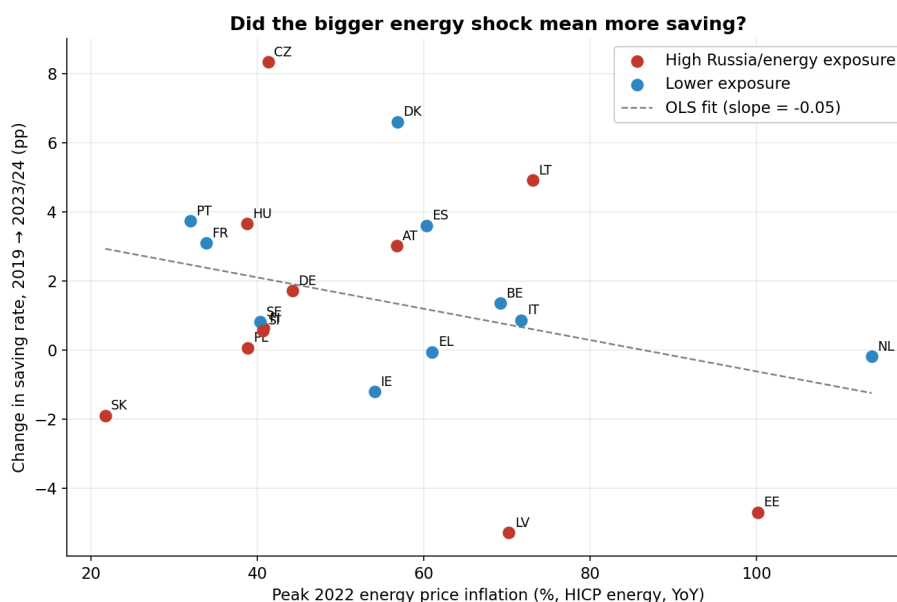


Figure 2: Leg B. Each country's peak 2022 energy inflation (the shock) against its change in saving 2019→2023/24. Colour marks editorial Russia/energy exposure; the dashed line is the OLS fit. The slope is positive but noisy.

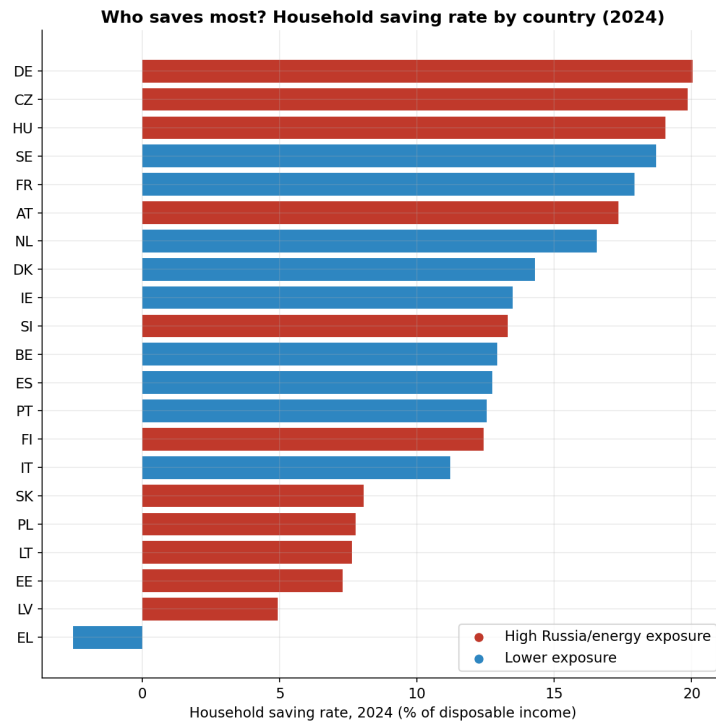


Figure 3: Leg B, levels. Household saving rate by country in the latest year. Germany and the Nordics sit at the top—but they were structurally high savers before the war, which is the confound that motivates using the *change* (previous figure) for any causal reading.

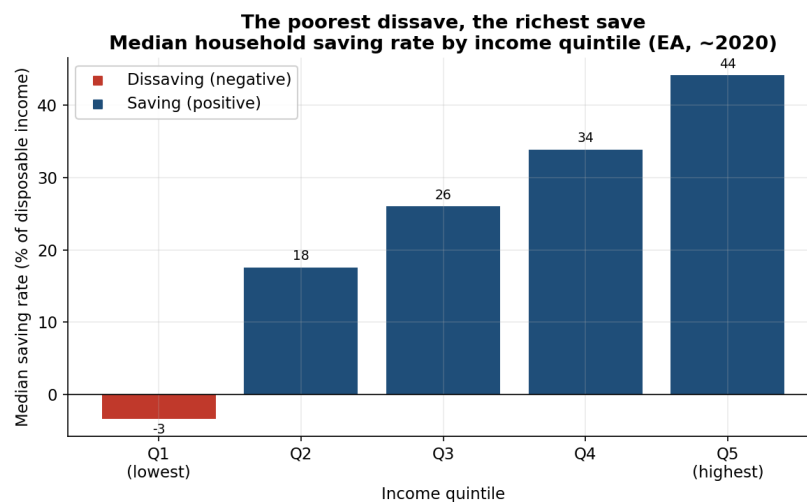


Figure 4: Leg C, structure. Median saving rate by income quintile (Eurostat ICW, latest snapshot). The bottom quintile *dissaves*; saving climbs steeply to the top. Any rise in the aggregate rate is mechanically a story about the upper part of the distribution.

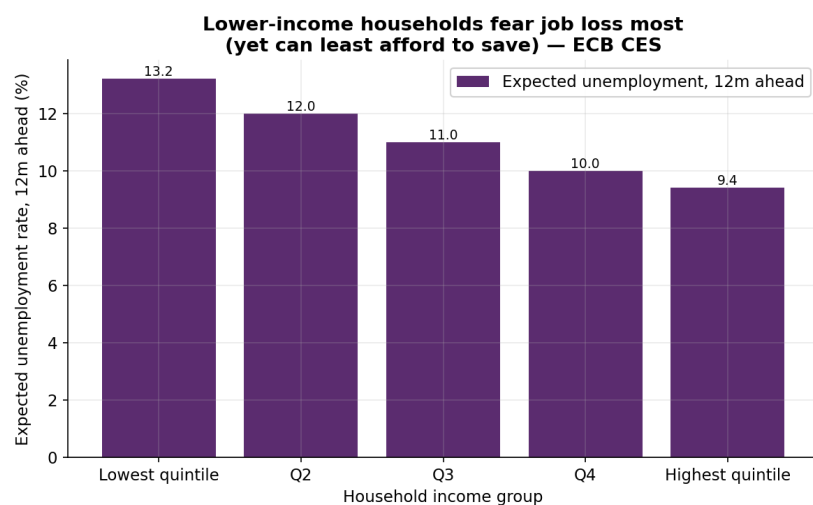


Figure 5: Leg C, fear. Expected unemployment 12 months ahead by income group (ECB CES). Lower-income households fear job loss most, yet (previous figure) can least afford to save—the equity twist of the precautionary story.

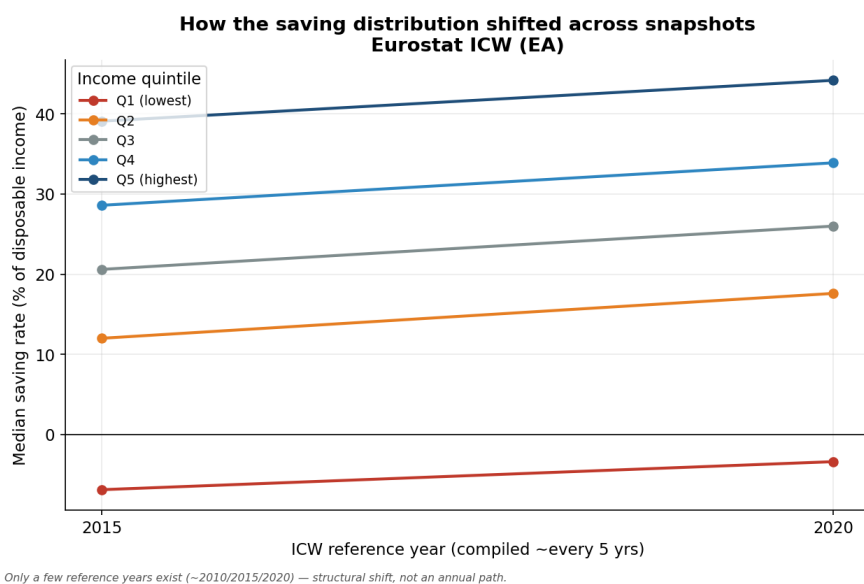


Figure 6: Leg C, over time. The saving rate by quintile across the few ICW reference years. Read as a structural shift between snapshots, not as an annual path.

5 Econometric methodology

This section is deliberately detailed. The aim is to test leg (A) —does uncertainty *drive* saving?— while taking seriously the statistical properties of the series. All tests run on the aligned quarterly panel $\{\text{saving}_t, \text{proxy}_t, \text{rate}_t, \text{inflation}_t\}, t = 2000\text{Q4} \dots 2025\text{Q4}$.

5.1 Design principle: respect the integration order

An earlier version of this analysis differenced *every* series and fitted a VECM. That was a mistake worth recording, because it motivates the current design:

- If a series is already stationary (I(0)), differencing it *over-differences*: it discards the level information that carries the precautionary signal and biases impulse responses toward zero (a unit-root/non-invertible MA component is introduced).
- A VECM (and the Johansen test behind it) is only meaningful when *all* series are I(1) and share cointegrating relationships. Feeding it stationary variables produces a “long-run” reading that is an artefact.

The current strategy therefore (1) classifies each series’ order, (2) tests the long-run relationship with a method valid for a *mix* of I(0) and I(1) regressors (ARDL bounds), and (3) runs Granger and the VAR/IRF in the representation each series actually needs—levels for the stationary core. ARMA is intentionally not used: it models a single series’ own memory, not a driver relationship.

5.2 Step 1 — Integration order from ADF and KPSS together

We classify each series with two complementary unit-root tests, because they have *opposite* null hypotheses and so cross-check each other.

Augmented Dickey–Fuller (ADF). Regress

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t, \quad H_0 : \gamma = 0 \text{ (unit root, non-stationary)}. \quad (1)$$

Lag length p is chosen by AIC. A *small* p -value rejects the unit root $\Rightarrow$ the series is stationary.

KPSS. Decompose $y_t = \zeta t + r_t + \varepsilon_t$ with a random walk $r_t = r_{t-1} + u_t$ and test

$$H_0 : \sigma_u^2 = 0 \text{ (level/trend stationary)}. \quad (2)$$

Here the null is the *opposite* of ADF’s. We use the level-stationary form (regression="c"). A *large* p -value fails to reject stationarity.

Decision rule. The series is confidently I(0) when ADF rejects *and* KPSS does not; confidently I(1) when ADF does not reject *and* KPSS does. When the two disagree (“CONFLICT”) the order is genuinely ambiguous—a hallmark of borderline or structurally-broken series—which is exactly the case the long-run test below is chosen to tolerate. The collector codes the order as I(0) iff ADF rejects at 5%, and flags agreement with KPSS.

Reading. The *core* variables—saving and the GPR proxy—are clearly I(0): both tests agree. The two controls are ambiguous (borderline I(1)). A system mixing I(0) and ambiguous-I(1) series is precisely where blanket differencing or Johansen would mislead, and where ARDL bounds is the right tool.

Table 1: Integration order, full sample (proxy = GPR). p -values; ADF H_0 : unit root, KPSS H_0 : stationarity.

series	ADF p (level)	KPSS p (level)	ADF p (Δ)	order	agree?
saving	0.017	0.087	0.000	I(0)	yes
gpr	0.000	0.100	0.000	I(0)	yes
rate	0.061	0.083	0.001	I(1)	CONFLICT
inflation	0.652	0.100	0.000	I(1)	CONFLICT

5.3 Step 2 — Long-run relationship: the ARDL bounds test

We test for a *level* (long-run) relationship between saving and the drivers with the Pesaran–Shin–Smith (2001) bounds test, estimated through the unrestricted (conditional) error-correction model. For dependent variable $y_t = \text{saving}_t$ and regressors $x_t = (\text{proxy}, \text{rate}, \text{inflation})$,

$$\Delta y_t = c + \sum_{i=1}^{p-1} \phi_i \Delta y_{t-i} + \sum_j \sum_{l=0}^{q_j} \theta_{jl} \Delta x_{j,t-l} + \underbrace{\rho y_{t-1} + \sum_j \pi_j x_{j,t-1}}_{\text{lagged levels}} + \varepsilon_t. \quad (3)$$

The bounds F -test has

$$H_0 : \rho = \pi_1 = \dots = \pi_k = 0 \quad (\text{no long-run level relationship}). \quad (4)$$

Why this test. Its decisive advantage is that the critical values come as a *pair of bounds*: a lower bound computed as if all regressors were I(0) and an upper bound as if all were I(1). The decision is therefore *order-agnostic*—valid for any mix of I(0) and I(1) regressors (it requires only that no series is I(2)):

- $F > \text{upper bound} \Rightarrow \text{reject } H_0$: a long-run relationship exists regardless of integration order;
- $F < \text{lower bound} \Rightarrow \text{cannot reject } H_0$: no long-run relationship;
- $\text{lower} \leq F \leq \text{upper} \Rightarrow \text{inconclusive}$.

Given the CONFLICT flags in Table 1, this is the correct long-run test; Johansen is reported only for reference (below) and should be ignored unless every series is I(1).

Implementation. The forced-saving COVID quarters (2020) are dropped as outliers. The model order is selected by minimising AIC over endogenous lags $L \in \{1, \dots, 4\}$ and exogenous order $O \in \{1, \dots, 4\}$ (every exogenous order ≥ 1 , as the UECM requires), with an unrestricted constant and no trend (Pesaran et al. “case III”).

Result (GPR). The AIC-selected specification is $L = 3, O = 1$. The bounds statistic is

$$F = 0.94 \quad \text{vs. the 5\% I(1) upper bound 4.01,}$$

so F lies *below even the lower bound*: we cannot reject H_0 . There is **no statistical evidence of a long-run level relationship** between the saving rate and the drivers over this sample. (For reference only, the Johansen trace implies rank 2, but with mixed integration orders this is uninformative and disregarded.)

Table 2: ARDL bounds critical values (Pesaran–Shin–Smith, case III). The test statistic $F = 0.94$ is below the lower bound at every level.

significance	lower bound, I(0)	upper bound, I(1)
10%	2.46	3.52
5%	2.88	4.01
1%	3.78	5.05

5.4 Step 3 — Granger causality

Granger causality asks whether past values of the proxy improve the prediction of saving *beyond* saving’s own past. We test it in the stationary representation of each series (levels for the I(0) core, here), drop COVID 2020, and scan lags 1–4:

$$\text{saving}_t = \alpha + \sum_{i=1}^m \beta_i \text{saving}_{t-i} + \sum_{i=1}^m \eta_i \text{proxy}_{t-i} + u_t, \quad H_0 : \eta_1 = \dots = \eta_m = 0. \quad (5)$$

An F -test $p < 0.05$ on the $\{\eta_i\}$ means the proxy Granger-causes saving.

Result (GPR, full sample). The p -values for “uncertainty $\rightarrow$ saving” are 0.91, 0.44, 0.71, 0.72 at lags 1–4: **no Granger causality** from uncertainty to saving at any lag. (The reverse direction, saving $\rightarrow$ uncertainty, is significant at lags 3–4, but a household saving rate “predicting” a global geopolitical-risk index is not economically meaningful and most likely reflects common low-frequency movement.) Two cautions apply: Granger causality is *predictive*, not structural; and on ~ 100 quarters with a large COVID hole the test has limited power, so a null is weak evidence of absence, not evidence of a null.

5.5 Step 4 — Structural VAR and impulse responses

To trace the dynamic response of saving to an uncertainty shock while controlling for the confounders, we estimate a VAR and compute orthogonalised impulse-response functions (IRFs).

Specification. With $Y_t = (\text{proxy}, \text{rate}, \text{inflation}, \text{saving})'$,

$$Y_t = c + \sum_{i=1}^p A_i Y_{t-i} + B D_t + u_t, \quad u_t \sim (0, \Sigma), \quad (6)$$

where D_t are dummies for the forced-saving COVID quarters (2020Q2–2021Q1) so that the pandemic spike does not contaminate the dynamics. Because the core series are I(0) (Table 1), the VAR is estimated *in levels*, and the IRF is read directly as a level response—no cumulation, no over-differencing. (If the core were not both I(0), the code estimates in first differences and cumulates the IRF back to levels.) Lag length p is chosen by AIC and capped at 4 to keep the responses stable on a short sample.

Identification. Shocks are orthogonalised by a Cholesky factorisation $\Sigma = PP'$ with the recursive ordering

$$\text{proxy} \rightarrow \text{rate} \rightarrow \text{inflation} \rightarrow \text{saving}.$$

The identifying assumption is that a variable ordered earlier may affect those ordered later *within the quarter*, but not the reverse. Placing uncertainty *first* treats it as the most exogenous block—a geopolitical-risk index is not plausibly driven by euro-area household saving within a quarter—and placing saving *last* lets it react contemporaneously to all three drivers. This is the natural ordering for the precautionary question; like any recursive scheme it is an assumption, not a fact, and is the main identification caveat.

Result (GPR). The response of saving to a +1 standard-deviation GPR shock is *positive but statistically insignificant*:

sample	peak response	95% CI at peak
full (2000Q4–2025Q4)	+0.108 pp at 0q	[−0.015, +0.230]
subsample (2010+)	+0.161 pp at 6q	[−0.019, +0.340]

The sign is *consistent with* precaution—more uncertainty, more saving—but the confidence interval includes zero in both samples, so the effect is not statistically distinguishable from no effect. Figure 7 plots the full-sample IRF with its band.

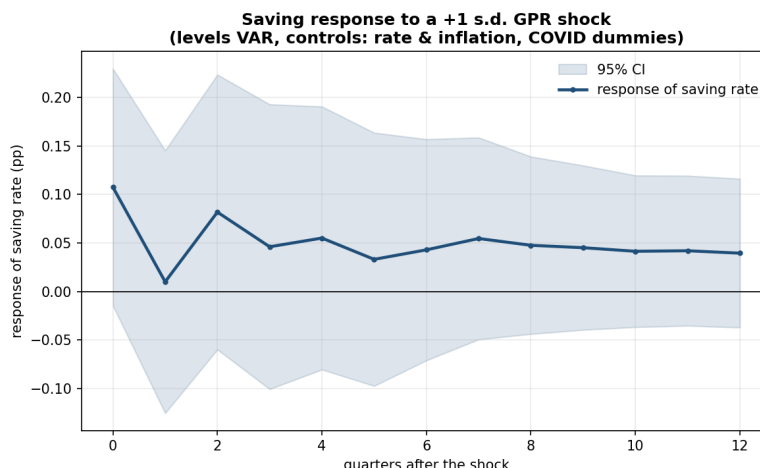


Figure 7: Orthogonalised impulse response of the saving rate to a +1 s.d. GPR shock (levels VAR; controls: ECB rate and HICP inflation; COVID dummies). The 95% band straddles zero throughout.

5.6 Step 5 — Robustness: subsample and proxy choice

Two robustness dimensions are built in. First, the key tests are re-run on a **post-2010 subsample** (64 quarters)—the period most relevant to the modern saving puzzle—with the same qualitative result (positive, insignificant IRF; no Granger causality). Second, the entire analysis can be re-run with the **EPU** proxy instead of GPR (-proxy epu); EPU yields the same narrative (a positive but insignificant response), confirming the conclusion is not an artefact of one proxy.

5.7 Step 6 — Proxy and timing: lead/lag cross-correlation

Finally we ask, descriptively, *which* proxy and at *what* lead/lag best tracks saving. After z-scoring both series we compute the cross-correlation at quarterly shifts $k = -8, \dots, +8$,

$$\rho_k = \text{corr}(z(\text{saving})_t, z(\text{proxy})_{t-k}), \quad (7)$$

where $k > 0$ means the proxy *leads* saving. We report it on both levels and first differences (Δ), because the levels of trending series mechanically inflate correlations, and we drop COVID 2020.

Result. In levels, GPR attains the larger peak correlation ($|\rho| = 0.42$) than EPU (0.35), which is why GPR is the preferred headline proxy. The peak occurs at $k = -4$ for GPR (i.e. the proxy is contemporaneous-to- slightly-lagging rather than cleanly leading), and in differences the peaks are weaker (-0.32 for GPR at $k = -2$). The lead/lag evidence is therefore suggestive of co-movement but does *not* pin down a clean “uncertainty-leads-saving” timing—consistent with the insignificant formal tests. Figure 8 shows the full profiles.

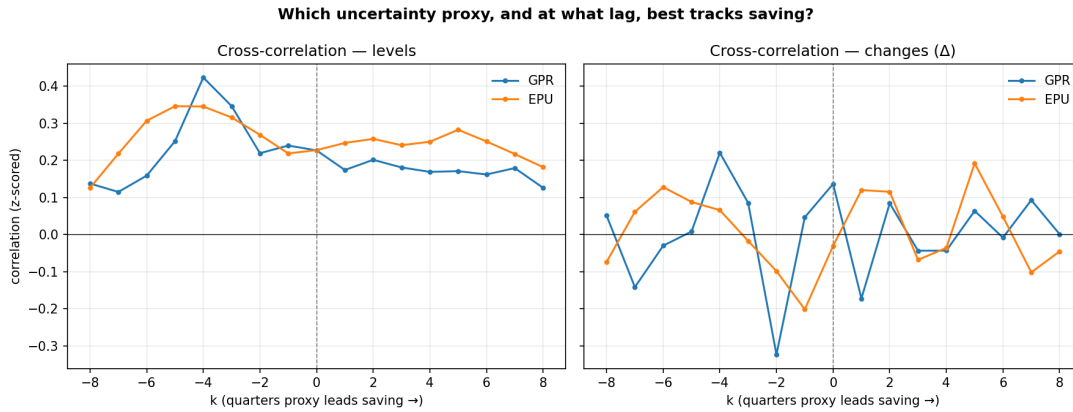


Figure 8: Lead/lag cross-correlation of the saving rate with each proxy, in levels (left) and changes (right). $k > 0$: proxy leads saving. COVID 2020 dropped.

6 Did rates or fear drive it? The follow-the-money test

The methodology of Section 5 targets the *level* of the saving rate and returns nulls: with so few quarters and three drivers moving together after 2022, no test can cleanly assign the rise to precaution. This section asks a different, sharper question — about the *composition* of saving rather than its level — where the precautionary and the “yield-chasing” (intertemporal-substitution) explanations make *opposite* predictions and can therefore be separated. The analysis lives in `extension_follow_money/`.

6.1 The discriminating idea: where the money goes

Fear and yield make opposite predictions about *where* households park new saving. Precaution favours liquid, instant-access assets (the buffer must be reachable in an emergency); yield-chasing favours higher-paying but less-liquid ones. We split households’ annual net acquisition of financial assets (Eurostat NASA_10_F_TR, sector S14_S15, euro area, current-price flows) into two buckets and track the *tilt* between them:

$$\begin{aligned} \text{instant-access} &= \text{currency (F21)} + \text{overnight deposits (F22)}, \\ \text{locked-for-yield} &= \text{time deposits (F29)} + \text{bonds (F3)}, \\ \text{tilt}_t &= \text{locked-for-yield}_t - \text{instant-access}_t \quad (\text{EUR bn/yr}). \end{aligned}$$

A positive tilt means that year’s saving leaned toward yield; negative, toward instant cash.

6.2 Linking the asset flows back to the saving rate

The tilt is a *composition* measure, so it does not map to the saving rate on its own. The bridge is an accounting identity: when a household saves, the money flows into financial assets, into housing, or into paying down debt; the part going into financial assets is exactly the *sum of all the asset-type flows we decompose*, and the tilt is *where within that sum* the money leans. So if the decomposition is legitimate, combining all the buckets should track the saving rate.

It does. Summing the asset flows and scaling by household gross disposable income ($B6G$, Eurostat NASA_10_NF_TR) gives a series that correlates $+0.87$ with the saving rate in levels and $+0.73$ as a share of income; both spike in 2020, dip in 2022–23, and recover. As a reconciliation, rebuilding the rate from the raw income accounts (gross saving $B8G$ over disposable income $B6G$) reproduces the published saving rate almost exactly (correlation $+1.00$; Table 3). The combined-flows line sits a few points *below* the saving rate — the wedge is mainly housing investment, which saving funds but is not a financial asset — yet the two clearly co-move (Figure 9). This places the composition analysis firmly inside the saving story: the tilt describes *how the saving in the headline rate was deployed*.

Table 3: Reconciliation: the rate rebuilt from the income accounts ($B8G/B6G$) matches the published saving rate, and the combined financial-asset flow tracks it (all in % of disposable income).

year	combined flows	saving rate ($B8G/B6G$)	saving rate (published)
2020	15.0	19.7	19.5
2022	8.8	13.6	13.5
2023	6.9	14.4	14.2
2025	9.3	15.0	14.8

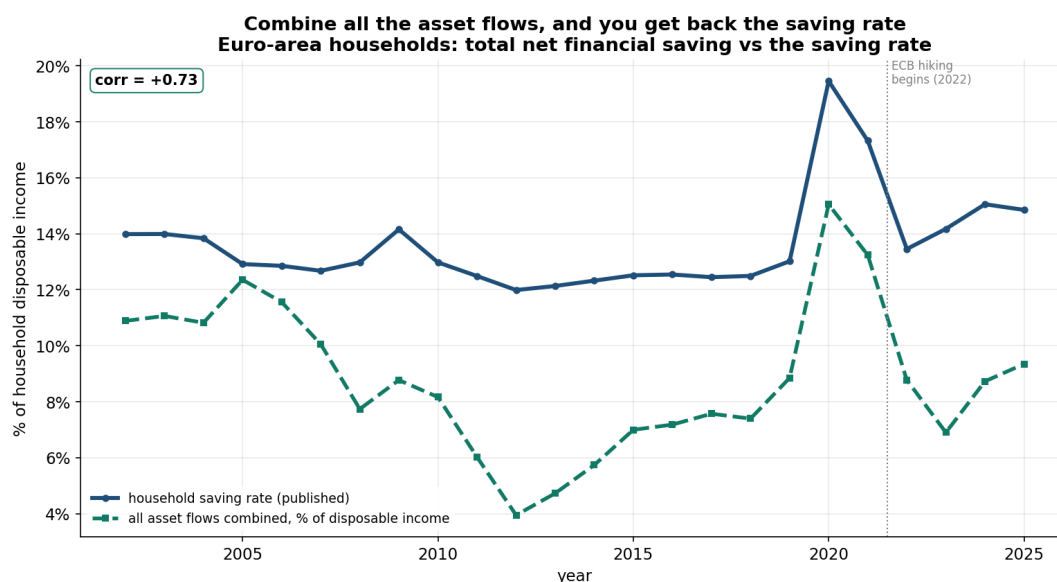


Figure 9: Combining all the asset flows reproduces the saving-rate path. Navy: the published household saving rate. Green dashed: the sum of all net financial-asset flows, as a share of disposable income. They co-move ($\text{corr} = +0.73$); the roughly constant gap is housing investment.

6.3 Descriptive evidence

The destination of saving flipped after the ECB began raising rates in mid-2022. The share of yearly financial saving going into instant-access cash fell from about 66% (2015–19) to about 2% (2022–25), while net household *bond* purchases went from negative (net selling of roughly EUR 40–50 bn/yr in 2019–21) to +EUR 96 bn (2022) and +EUR 311 bn (2023). Households went from shunning bonds to buying them in record size precisely as yields rose — the reach-for-yield pattern, and the opposite of the 2020 COVID episode, when saving piled into instant-access cash (the precautionary pattern). The tilt correlates +0.79 with the ECB policy rate (annual). Figures 10 and 11 show the picture.

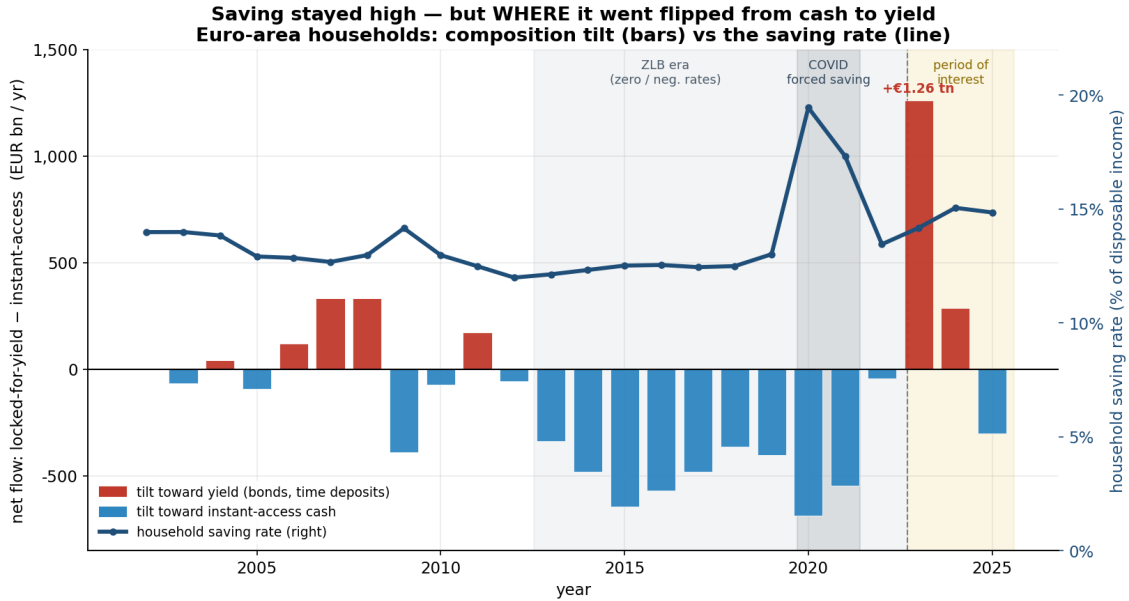


Figure 10: Composition tilt (bars; locked-for-yield minus instant-access, EUR bn/yr) against the household saving rate (line). The saving *level* stays high throughout — peaking in the COVID year, when the tilt is most negative (cash) — while the *composition* flips to yield (red) once rates rise. Regimes shaded as in chart A: the zero-lower-bound era, COVID forced saving, and the post-2022 period of interest.

6.4 A model comparison: rates/inflation versus precaution

Because the tilt is signed, the two stories give opposite, testable predictions. We regress the tilt on the ECB rate, inflation, and an uncertainty proxy (the Geopolitical Risk index, GPR), each standardised to a z-score so the coefficients are comparable (the effect of a one-standard-deviation move, in EUR bn):

$$\text{tilt}_t = \beta_0 + \beta_1 \tilde{r}_t + \beta_2 \tilde{\pi}_t + \beta_3 \tilde{u}_t + \varepsilon_t, \quad (8)$$

with $\tilde{r}$ the rate, $\tilde{\pi}$ inflation, $\tilde{u}$ uncertainty. The competing hypotheses are sharp:

$$\text{yield-chasing} \Rightarrow \beta_1 > 0, \quad \text{precaution} \Rightarrow \beta_3 < 0$$

(more uncertainty should push saving *toward* instant cash, a more negative tilt). Standard errors are heteroskedasticity-robust (HC3), appropriate for the small sample.

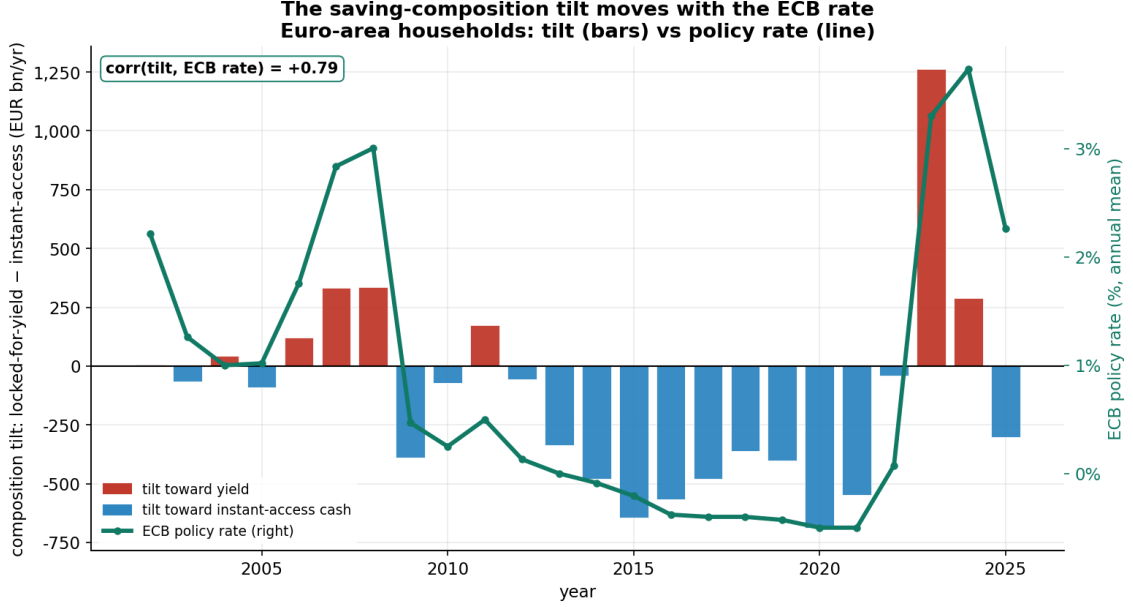


Figure 11: The composition tilt (bars) against the ECB policy rate (line); the two move together, $\text{corr} = +0.79$ over 2002–2025.

How the models are ranked. The “rates+inflation” and “precaution” models are *non-nested* — neither is a special case of the other — so they cannot be compared with an ordinary F -test. We use the Akaike Information Criterion,

$$\text{AIC}_m = 2k_m - 2 \ln \hat{L}_m, \quad (9)$$

where k_m is the number of parameters and $\hat{L}_m$ the maximised likelihood of model m ; lower is better (it rewards fit and penalises extra parameters). Raw AIC values are not interpretable alone, so we convert them to *Akaike weights*,

$$\Delta_m = \text{AIC}_m - \min_j \text{AIC}_j, \quad w_m = \frac{\exp(-\frac{1}{2}\Delta_m)}{\sum_j \exp(-\frac{1}{2}\Delta_j)}. \quad (10)$$

Each w_m is the probability that model m is the best (Kullback–Leibler) model in the candidate set — exactly the “how much more likely is rates/inflation than precaution?” statement we want.

6.5 Results

Table 4 reports the three models for the tilt (annual levels, $n = 24$).

The reading is unambiguous. The ECB rate is the dominant driver: a one-standard-deviation rate increase shifts the tilt by roughly +EUR 280–300 bn toward yield, significant at 5%. The precaution model is almost empty (adjusted $R^2 \approx 0.02$), and its uncertainty coefficient has the *wrong* sign for precaution and is insignificant. Head-to-head, the Akaike weights put essentially 100% on rates+inflation versus 0% on precaution. Using net bond purchases as the dependent variable gives the same verdict (rate coefficient +, $p = 0.01$; 100% vs. 0%); there, uncertainty looks significant *on its own* but collapses to zero once the rate is included — a textbook confounder, with uncertainty merely standing in for rates.

Table 4: Composition tilt on standardised drivers (annual, 2002–2025, $n = 24$; HC3 robust SEs). Coefficients in EUR bn per one-standard-deviation of the driver; * denotes significance at 5%. Akaike weights are computed across the three models. Inflation enters positively but is insignificant throughout.

model	adj. R^2	AIC	rate (β_1)	uncertainty (β_3)	Akaike weight
rates + inflation	0.74	329.8	+278*	—	25%
precaution (GPR)	0.02	360.5	—	+103	$\approx 0\%$
all three	0.77	327.5	+304*	−96	75%
head-to-head (rates+inflation vs. precaution): $\approx 100\%$ vs. $\approx 0\%$					

6.6 Robustness and scope

Regressing levels of trending series can manufacture significance, so we re-estimate (8) in first differences. The rate coefficient keeps the right (positive) sign but loses significance ($p \approx 0.13$) on $n = 23$ noisy annual changes; the model *ranking* is nonetheless unchanged — precaution still receives essentially zero Akaike weight. Two caveats bound the claim. The sample is small (~ 24 annual euro-area observations) and the 2023 bond surge is a high-leverage point; robust errors and the differenced re-run mitigate but do not remove this. And, most important, this test concerns the *composition* of saving — *how* households deployed it — not the *level* of the saving rate: it shows the reallocation of saving is far more likely a reaction to rates and inflation than to precaution, but does not by itself explain why the aggregate rate rose, which stays overdetermined (Section 5).

7 Interpretation

Putting the legs together:

- **Leg A (time series).** Descriptively strong: saving is elevated and rose alongside uncertainty after 2022 (Fig. 1). But once the ECB rate and inflation are controlled for, the formal tests cannot establish a precautionary channel: no long-run relationship (ARDL $F = 0.94$), no Granger causality, and a positive-but-insignificant impulse response. The honest conclusion is *co-movement consistent with, but not proof of, precaution*.
- **Leg B (cross-country).** Suggestive and confounded: the slope of saving-change on energy-shock is positive but noisy, and the level map mostly reflects long-standing North–South habits.
- **Leg C (distribution).** The clearest structural fact: the bottom quintile dissaves and the top saves, so any aggregate rise is concentrated among households with slack—while lower-income households report the most fear. The same uncertainty produces a bigger buffer at the top and forgone consumption at the bottom.

The overall verdict is intentionally measured: the precautionary story is *plausible and consistent with the data*, but the time-series evidence is not strong enough to claim a clean causal channel—which is why the descriptive charts and the econometrics are presented together rather than one standing in for the other. On the narrower question of *how* households saved, however, the evidence is sharper: the follow-the-money test (Section 6) finds the post-2022 reallocation toward yield is far more likely a reaction to rates and inflation than to precaution — a statement the level data alone cannot support.

8 Limitations and caveats

- **Power.** The aligned sample is short (~ 101 quarters) and contains a large COVID structural break; time-series tests on such samples have low power, so null results are weak evidence of absence.
- **Identification.** The IRF rests on a recursive (Cholesky) ordering; a different ordering or a sign-restriction scheme could shift the point estimates (though the wide bands suggest not the significance).
- **Proxies.** Uncertainty is latent; GPR and EPU are imperfect, newspaper-based measures.
- **Gross vs. net saving.** The headline target is the *gross* household saving rate; cross-source comparisons should note this.
- **Distributional timeliness.** ICW is experimental and only ~ 5 -yearly; it shows structure, not the post-2022 dynamics. The OECD annual panel that would show year-by-year movement is not currently retrievable from the queried flow.
- **Editorial exposure grouping.** The red/blue Russia-energy split in leg B is a simplification; the continuous energy-inflation axis is the rigorous version.
- **Composition vs. level.** The follow-the-money test (Section 6) speaks to *how* saving was allocated, not *why* the aggregate rate rose, and rests on ~ 24 annual observations with a high-leverage 2023.
- **Vintages.** All series are pulled live and revised by providers; regenerate before citing any number.

A Reproduction details

A.1 File tree

```
01_collect_data.py      # STEP 1  data collection -> data/*.csv
02_make_figures.py      # STEP 2  figures          -> figures/*.png
03_econometrics.py      # STEP 3  econometrics        -> results.md, A2, A3
run_all.sh              # pip install + run 01->02->03 (classic Python, no venv)
requirements.txt         # pinned dependencies
data/                   # generated tidy CSVs
figures/                 # generated charts
extensions/              # identification extensions (panel FE, event study, ...)
extension_follow_money/ # follow-the-money composition tests + econometrics (Sec. 6)
findings.md              # narrative write-up of the results
documentation.tex        # this document
legacy/                  # superseded original scripts (safe to delete)
```

A.2 Commands

```
# one-shot, fully reproducible
bash run_all.sh

# or stage by stage (after: pip install -r requirements.txt)
python 01_collect_data.py
```

```
python 02_make_figures.py
python 03_econometrics.py          # GPR (default)
python 03_econometrics.py --proxy epu --substart 2010
```

A.3 Data dictionary (generated CSVs)

file	frequency	contents
ea_saving_rate_quarterly.csv	quarterly	date, saving (% GDI)
gpr.csv	monthly	date, GPR index
fred_eu_epu.csv	monthly	date, EU EPU index
ecb_rate.csv	daily	date, ECB deposit rate (%)
ea_inflation.csv	monthly	date, HICP YoY (%)
country_saving_annual.csv	annual	geo × year saving pivot
country_energy_peak_2022.csv	2022 scalar	geo, peak energy infl. (%)
scatter_energy_vs_saving.csv	cross-section	geo, saving rise, energy peak
saving_rate_by_quintile.csv	snapshot	q, value, geo, year
icw_quintile_panel.csv	snapshots	year, q, value, geo
ces_unemp_expectations.csv	latest	group, expected unemployment (%)
econometrics_results.md	—	full text log of step 3

All numerical results in this document come from a clean end-to-end run and are reproducible via `run_all.sh`. Because the underlying series are revised by their providers, regenerate the data and re-compile before citing.